

Represent 4-Digit Numbers



Be Curious

What do you notice?
What do you wonder?

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Math is... **Mindset**

How do you feel about
learning math?

Learn

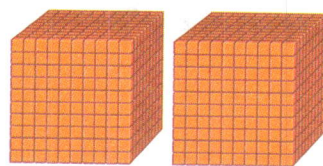
What are some ways to represent this number?



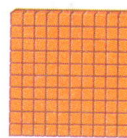
You can use a place-value chart.

thousands	hundreds	tens	ones
2	1	3	8

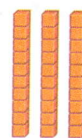
You can use base-ten blocks.



2 cubes
2,000



1 flat
100



3 rods
30



8 units
8

You can read and write the number in different forms.

standard form 2,138

expanded form $2,000 + 100 + 30 + 8$

word form two thousand, one hundred thirty-eight

Math is... Generalizations

How is representing 4-digit numbers the same as representing 3-digit numbers?

4-digit numbers have thousands, hundreds, tens, and ones.

Work Together

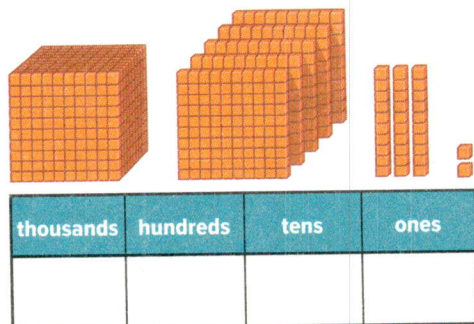
How can you represent 1,208 in expanded form and word form?

On My Own

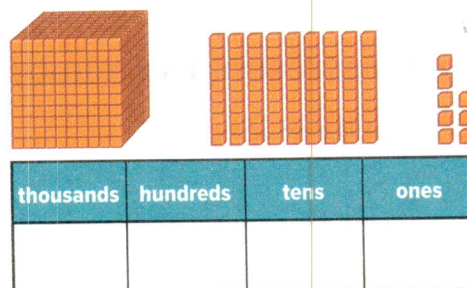
Name _____

What number is represented by the base-ten blocks?

1.



2.



How can you represent the number in the place-value chart and in expanded form?

3. 2,446

thousands	hundreds	tens	ones

4. 4,729

thousands	hundreds	tens	ones

How can you represent the number shown in standard form and expanded form?

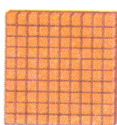
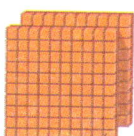
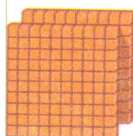
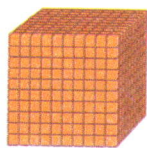
5. three thousand, one hundred twelve

6. six thousand, eighty-seven

7. seven thousand, two hundred twenty-four

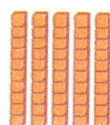
How can you represent the number shown in standard form and expanded form?

8.



Tens

Ones



1,553

How can you represent the number in standard form?

9. $8,000 + 500 + 2$

10. $9,000 + 50 + 2$

11. **STEM Connection** 1,455 customers visited Saffron's pastry shop this month. Last month 1,355 customers came to the shop. Explain how she can use place value to determine the difference in the number of customers.



12. **Extend Your Thinking** Use the digits shown to write a number with the least possible value. Write the number in standard form, expanded form, and word form.

3

8

8

2

Reflect

How does place value help you write a number in word form?

Math is... **Mindset**

What feelings did you experience about learning math?

Round Multi-Digit Numbers



Be Curious

What do you notice?
What do you wonder?



Math is... **Mindset**

What are some ways you can connect with your classmates?

Learn

About how many blocks are there?

You can round to describe about how many.

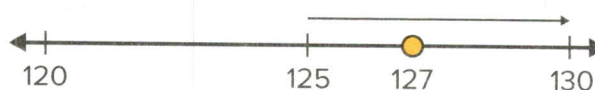
When you **round** to the nearest 10 or 100, you determine a ten or a hundred that is close to the original number.



► **One Way** Use a number line to round.

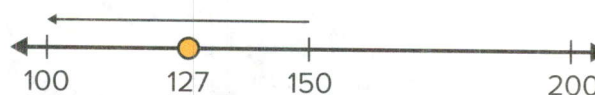
Round to the nearest 10.

127 is to the right of the halfway point. It rounds to 130.



Round to the nearest 100.

127 is to the left of the halfway point. It rounds to 100.



► **Another Way** Use place value to round.

Round to the nearest 10.

127

If the digit in the ones place is 5 or greater, round up.

130

Round to the nearest 100.

127

If the digit in the tens place is less than 5, round down.

100

You can round to make a number easier to work with when an exact number is not needed.

Math is... Choosing Tools

Why is a number line helpful for rounding?

Work Together

Ellie rounds 255 to 260. Carter rounds 255 to 300. Why are their rounded numbers different?

On My Own

Name _____

Use a number line to round.

1. Round 27 to the nearest 10.

2. Round 896 to the nearest 10.



Use place value to round.

3. Round 48 to the nearest 10.

4. Round 273 to the nearest 10.

Use a number line to round. Show your work.

5. Round 436 to the nearest 100.

6. Round 672 to the nearest 100.

436 rounded to the nearest 100
is _____.

672 rounded to the nearest 100
is _____.

7. How can the number 78 round to 80 and 100? Explain.

8. A number rounded to the nearest 10 is 240. What number could it be? _____

9. How can you use the number line to show why 678 rounded to the nearest 100 is 700? Explain your reasoning.



10. **Error Analysis** Tess says that 315 rounded to the nearest 10 is 310. Do you agree? Explain your reasoning.

11. Jack is making a fruit salad. He puts in 9 strawberries, 25 orange wedges, 19 kiwi slices, 27 blueberries, 16 grapes, and 21 raspberries. Which fruits does he use about 20 of?

12. **Extend Your Thinking** Sam is shopping. He has \$50. He wants to buy dog food that costs \$15, vegetables that cost \$22, and fruits that cost \$12. How can Sam make sure he has enough money?

Reflect

When might you want to round to the nearest 10 instead of the nearest 100?

Math is... Mindset

How did you connect with your classmates?

Rounding Numbers

Name _____

Circle *all* correct answers for the problem.

1. Suppose you rounded to the nearest 10. Which numbers below would round to 630? Circle *all* of them.

- a. 632
- b. 638
- c. 627
- d. 625
- e. 623
- f. 635
- g. 534
- h. 529

Explain your choices.

Circle *all* correct answers for the problem.

2. Suppose you rounded to the nearest 100. Which numbers below would round to 900? Circle *all* of them.

- a. 956
- b. 871
- c. 943
- d. 839
- e. 962
- f. 819
- g. 988
- h. 925

Explain your choices.

Reflect On Your Learning

I am
confused.

I'm still
learning.

I understand.

I can teach
someone else.

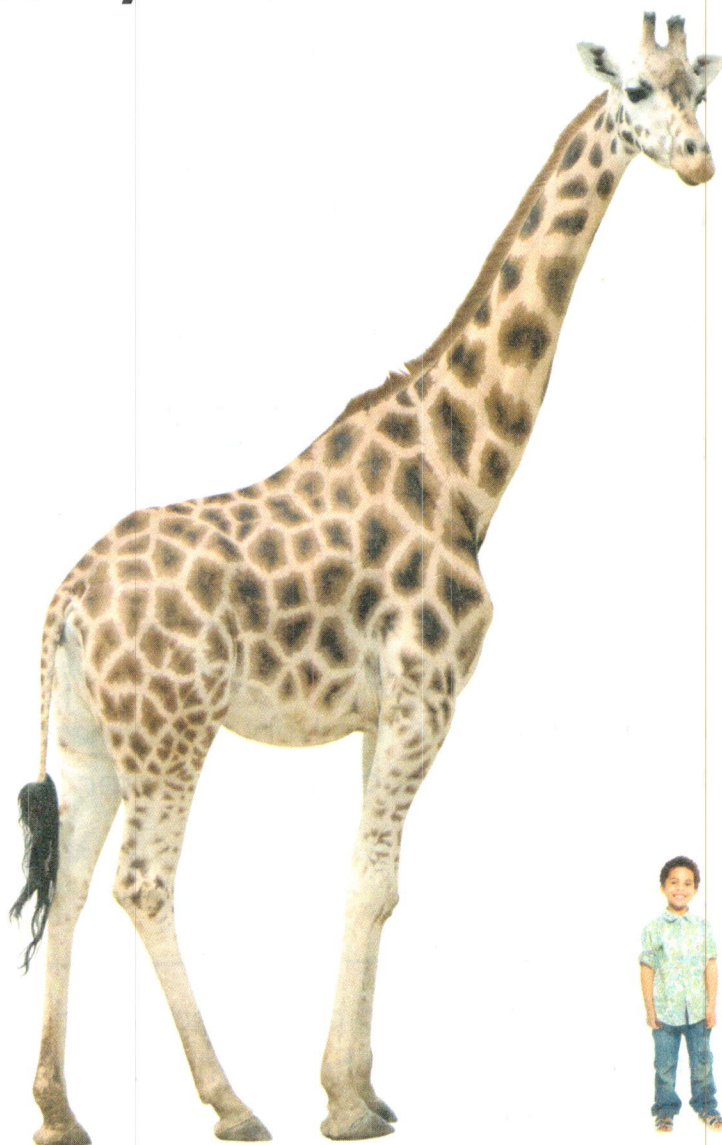


Estimate Sums and Differences



Be Curious

What could you ask?



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Math is... Mindset

How can you recognize and understand how others are feeling?

Learn

Carter is 122 centimeters tall. The giraffe is 576 centimeters tall.

About how much taller is the giraffe?

You can **estimate** the difference. An estimate is close to the exact answer.



► **One Way** Round the numbers to estimate the difference.

$$576 - 122 = ?$$

$$\begin{array}{r} \downarrow \quad \downarrow \\ 580 - 120 = 460 \end{array}$$

$$576 - 122 = ?$$

$$\begin{array}{r} \downarrow \quad \downarrow \\ 600 - 100 = 500 \end{array}$$

► **Another Way** Use **compatible numbers** to find the difference.

Compatible numbers are easy to work with.

$$576 - 122 = ?$$

$$\begin{array}{r} \downarrow \quad \downarrow \\ 575 - 125 = 450 \end{array}$$

Math is... Choosing Tools

How can estimating a sum or difference help to detect possible errors?

You can use rounding or compatible numbers to **estimate** an answer when you do not need an exact sum or difference.

Work Together

An elephant eats 428 pounds of food on Thursday and 503 pounds of food on Friday. About how many pounds of food does the elephant eat in the two days?

On My Own

Name _____

**What is a reasonable estimate of the sum or difference?
Write or draw to show your thinking.**

1. $423 + 168 = ?$

2. $? = 695 - 205$

3. $? = 317 + 248$

4. $473 + 218 = ?$

5. $798 - 307 = ?$

6. $? = 835 - 466$

7. How can you use rounding to estimate the sum of $389 + 223$?

8. A complete set of baseball cards has 678 cards. Julio needs 273 more cards to complete his set. How can you use rounding to find about how many cards are in Julio's collection?

9. The Comic Book Shack displays 318 comic books near the front door and keeps 502 comic books in the storage room. How can you use compatible numbers to find about how many comic books are in the store?

10. **STEM Connection** Saffron's bakery needs to decorate 355 cupcakes for an event. It has decorated 223 so far. How can she determine about how many more cupcakes they need to decorate?



11. Melinda estimates she traveled 830 miles last Monday and Tuesday. She traveled 412 miles on Tuesday. About how many miles could she have traveled on Monday?

12. **Extend Your Thinking** Jason has 744 flyers to deliver. If he has delivered 62 flyers at each of his last 2 stops, about how many flyers does he have left to deliver?

Reflect

When might you need to estimate a sum or difference in your life?

Math is... Mindset

How have you worked to understand how others are feeling?

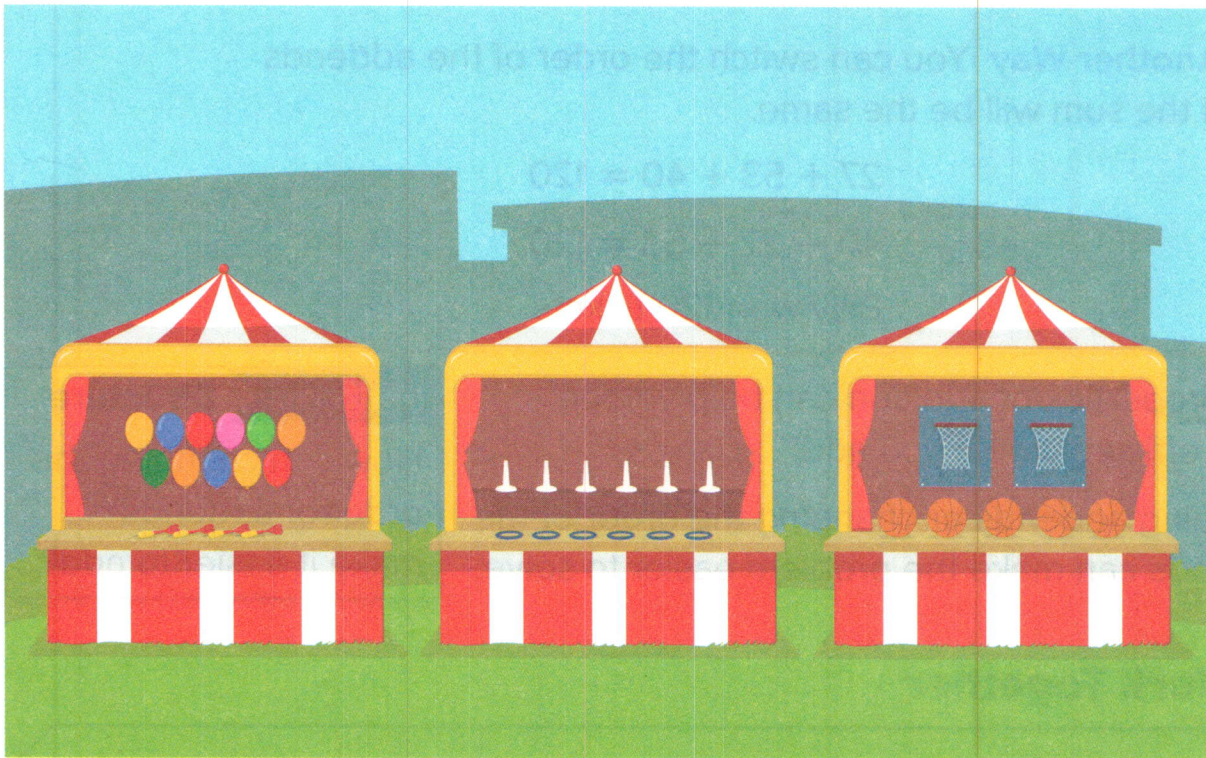
Use Addition Properties to Add



Be Curious

What question could you ask?

At the school carnival, Billy won some tickets at balloon darts, more tickets at ring toss, and the most tickets at basketball toss.



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Math is... Mindset

What do you do to avoid getting distracted?

Learn

At the school carnival, Billy won 27 tickets at balloon darts, 40 tickets at ring toss, and 53 tickets at basketball toss.



TICKET

How many tickets did he win?

► **One Way** You can group addends in any way and the sum will be the same.

$$\begin{array}{c} 27 + 53 + 40 \\ \swarrow \quad \searrow \\ 80 + 40 = 120 \end{array}$$

Add 53 and 40 first.

$$\begin{array}{c} 27 + 53 + 40 \\ \swarrow \quad \searrow \\ 27 + 93 = 120 \end{array}$$

► **Another Way** You can switch the order of the addends and the sum will be the same.

$$27 + 53 + 40 = 120$$

$$40 + 27 + 53 = 120$$

Billy won 120 tickets.

The properties of addition can help you be more efficient when you add.

Grouping addends or changing the order of the addends can make the problem easier to solve.

Math is... **Structure**

How can changing the order of addends make it easier to add?

Work Together

Josh is completing the equation $797 + 111 = \underline{\quad} + 797$. Josh adds $797 + 111$ and subtracts 797 to complete the equation. How can Josh complete the equation more efficiently?

On My Own

Name _____

How can you make the equation true?

1. $218 + 325 = 325 + \underline{\hspace{2cm}}$

2. $465 + \underline{\hspace{2cm}} = 78 + 465$

3. $529 + 407 = \underline{\hspace{2cm}} + 529$

4. $\underline{\hspace{2cm}} + 93 = 93 + 505$

5. Mauricio had a sale. The table shows the number of items he sold each day. Which expressions show how to find the total number of items Mauricio sold? Choose all that apply.

A. $42 + 67 + 58$

B. $67 - 58 + 42$

C. $58 + 42 + 67$

D. $58 + 67 + 24$

Items Sold	
Monday	58
Tuesday	67
Wednesday	42

6. How can you group the addends to make it easier find the sum? Explain your thinking.

$$372 + 264 + 228$$

How can you show one way to group the addends and solve?

7. $157 + 17 + 783$

8. $198 + 502 + 155$

9. $135 + 458 + 42$

10. $235 + 105 + 317$

11. **STEM Connection** Saffron needs to order the items listed. How can she group these numbers to find her total cost?

flour	\$367
boxes	\$358
cupcake liners	\$142

12. **Extend Your Thinking** Mrs. Ruiz is checking her receipt. The three items cost \$305, \$350, and \$195. How can she use properties of addition to add more efficiently?

Reflect

How can changing the order of three addends help you add more efficiently?

Math is... **Mindset**

How did you avoid getting distracted?

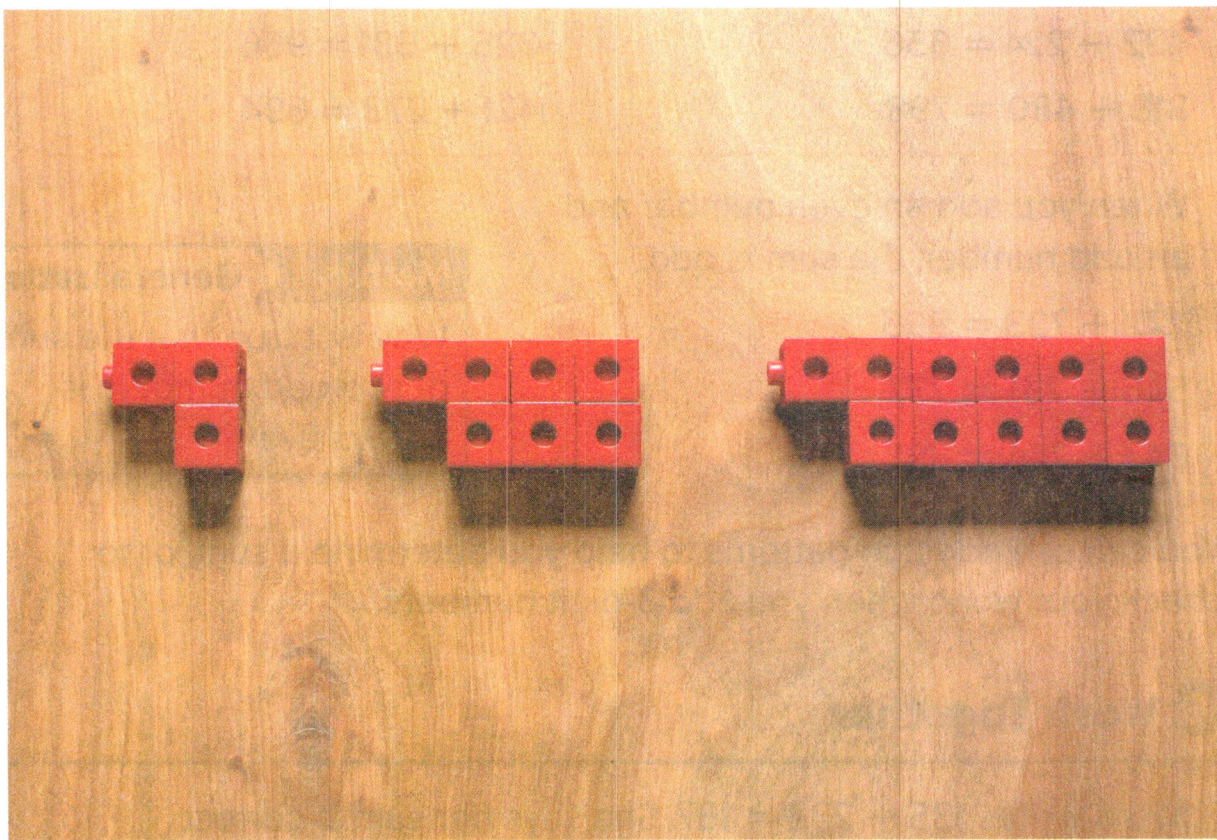
Addition Patterns



Be Curious

Is it always true?

Mena states that the total will always be odd if she continues to add 4 blocks to the set.



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Math is... Mindset

What strategies help you work more efficiently?

Learn

Mena notices that when she adds an even number and an odd number, the sum is odd.

How can she determine if the sum of an even number and an odd number is always odd?

When you add **even numbers** and **odd numbers**, there are patterns in the sums.

	$72 + 13 = 85$
	$18 + 41 = 59$
	$30 + 25 = 55$

When you add two even numbers, the sum is even.

$$246 + 100 = 346$$

$$432 + 224 = 656$$

$$318 + 480 = 798$$

When you add two odd numbers, the sum is even.

$$547 + 155 = 702$$

$$325 + 631 = 956$$

$$421 + 273 = 694$$

When you add an even number and an odd number, the sum is odd.

$$272 + 723 = 995$$

$$546 + 231 = 777$$

$$647 + 244 = 891$$

Math is... Generalizations

Why is it true that the sum of two odd numbers is always even?

You can use addition patterns to help you determine a sum, or to check your work, when you add 3-digit numbers.

Work Together

Nisha writes $135 + 232 = 167$. She says her sum is correct because an odd number added to an even number equals an odd sum. Do you agree with her reasoning? Explain.

On My Own



Name _____

What makes the equation true? Write *even* or *odd*. Then write two equations with 3-digit numbers to support your answer.

1. _____ = even + even

2. odd + even = _____

3. odd + _____ = even

4. even + odd = _____

5. even + _____ = even

6. _____ = odd + odd

What is the sum? Use patterns to help justify your answer.

7. $486 + 123 =$ _____

8. $154 + 272 =$ _____